Jack Mondry

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Summary

Principal Systems Engineer with 12+ years in medical device development, specializing in surgical robotics. Proven leader in cross-functional team management, systems engineering methods, and FDA-regulated product innovation. Expert in requirements management, risk assessment, and strategic development.

Core Competencies

• Requirements Development • System Architecture • Interface Definition • Risk Management (ISO 14971) • Usability & Human Factors (IEC 62366) • Design Controls (21 CFR 820) • Robotics • Cross-Functional Leadership • Verification & Validation • Clinical / Regulatory Collaboration

Experience

Perceptive Technologies / Cyberdontics (Boston, MA)

Robotics and AI imaging platforms for dental procedures and diagnostics

Principal Systems Engineer (July 2022 – present)

- Owned full system architecture and requirements definition for a dental surgical robot achieving sub-millimeter accuracy and FIH trial success.
- Developed usability engineering and risk management files
- Assessed and selected database for managing requirements, risks, and tests (MatrixALM) along with being key stakeholder in selection of eQMS (Propel PML) and related SOPs
- Generated content and participated in calls with FDA related to early feasibility studies and submission strategies
- Worked closely with external partners for software and electrical development, biocompatibility, reprocessing, clinical, and registered test laboratories to ensure successful delivery

Sharp End Systems, LLC. (Minneapolis, MN)

Principal (January 2021 – August 2022)

- Provided systems engineering consulting services for early-stage medical technology companies with clients in the fields of neurosurgery and obstetrics

Monteris Medical (Plymouth, MN)

MRI-guided, robotically controlled laser ablation device used in neurosurgery for oncology and epilepsy

Senior Systems Engineer (July 2019 – July 2022)

- Implemented systems engineering methodologies and influenced change across the organization to align with industry best practices
- Led team on development of a novel ceramic cranial access port and delivery system to effectively eliminate MR image artifacts
- Collaborated with surgeons and key stakeholders to define user needs, uncover unmet clinical challenges, and prioritize opportunities for new product concepts
- Managed and coached direct reports, Test Engineers and Electrical Technician

Senior Hardware Engineer (February 2018 – July 2019)

- Attended clinical cases, conferences, and customer meetings to bridge the gap between engineering and clinical use
- Supported post-market complaint investigations and critical CAPA closures

Minnetronix Medical (St. Paul, MN)

Extracorporeal filtration device for cerebral spinal fluid towards the treatment of conditions such as subarachnoid hemorrhage and meningitis

Senior Research & Development Engineer (June 2016 – February 2018)

- Developed device hardware leading to a successful IDE submission
- Worked with external partner to develop a dual-lumen spinal catheter
- Functioned as an on-site field clinical engineer for first-in-human trial and trained clinical staff
- Responsible for analysis of novel filtration technology and preparing technical approach for SBIR grants and creation of intellectual property disclosures with technology roadmap

Mechanical Engineer II (January 2013 – June 2016)

- Performed mechanical design for Class II and III electro-mechanical medical devices including neuro critical care, cardiac assist devices, nebulizers, and ablation systems

University of Nebraska-Lincoln (Lincoln, NE)

Graduate Research Assistant – Surgical Robotics Laboratory (August 2010 – December 2012)

- Designed and built a miniature surgical robot for single-port surgery leading to successful pre-clinical study with technology currently licensed by Virtual Incision ($146M raised)

NASA Jet Propulsion Laboratory (Pasadena, CA)

Engineering Intern – Nano and Micro Systems Group (June 2011 – August 2011)

- Designed mechanical linkage for an articulating stereoscopic (3D) endoscope for minimally invasive neurosurgery

Education

University of Nebraska-Lincoln

Master of Science in Mechanical Engineering (August 2012)

Bachelor of Science in Mechanical Engineering (May 2010)

- Minor in Business Administration

Professional Certifications & Honors

- Named inventor on 10+ US Patents issued or filed
- Product Manager Nanodegree (Udacity – April 2020)
- AAMI Human Factors for Medical Devices (September 2018)
- Advanced Cardiac Physiology and Anatomy (University of Minnesota – January 2014)
- Certified SolidWorks Professional, CSWP (August 2014)
- Engineer-in-Training (EIT) Certification (October 2009)